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Door flag

Abstract

A flag mechanism for an aircraft door includes a first arm and a second arm each having a flag on a first end and a catch on a second end. A first cam is operatively coupled to a first latching member and a second cam is operatively coupled a second latching member. The first and second latching members are each configured for latching the aircraft door, which are locked when the first and second cams are rotated to a locked position. A first stud extends from a side of the first cam and a second stud extends from a side of the second cam. The first and second studs are each configured to be received by a respective catch of the first and second arms while the first and second cams are rotated, thereby moving each respective flag from a first position to a second position for simultaneous visual inspection.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority of U.S. Provisional Patent Application No. 63/353,874, entitled Crew Door Flag and filed on Jun. 21, 2022, the disclosure of which is herein incorporated by reference in its entirety.

BACKGROUND
1. Field
(1) Embodiments of the invention relate generally to doors, and more specifically to doors having locking mechanisms.
2. Related Art
(2) Various solutions to door locking mechanisms have been disclosed in the art. For example, U.S. Patent Application Publication No. 2021/0229792 to Blum et al. discloses an actuating system that securely locks an actuatable door. U.S. Patent Application Publication No. 2019/0017304 to Bellavia et al. discloses a keeper for a lock of an aircraft door. U.S. Pat. No. 5,180,121 to Banks et al. discloses an aircraft door assembly. U.S. Patent Application Publication No. 2020/0181943 to Savidge et al. discloses an escape hatch of an aircraft. U.S. Pat. No. 10,746,378 to Bachman et al. discloses an optical display for displaying a mechanical state of a lock. U.S. Pat. No. 10,711,487 to Ward discloses a cabin door that includes a lock. Various other door locking mechanisms are disclosed in U.S. Pat. No. 8,752,794 to Gorgoglione, U.S. Patent Application Publication No. 2021/0070416 to Buchet, and U.S. Patent Application Publication No. 2020/0071978 to Holtrup et al.
SUMMARY
(3) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other aspects and advantages of the invention will be apparent from the following detailed description of the embodiments and the accompanying drawing figures.
(4) In an embodiment, a flag mechanism for a door of an aircraft includes: an arm comprising a flag disposed on a first end and a catch disposed on a second end; a rotatable arcuate member operatively coupled to a latching member, wherein the latching member is configured to be inserted into a frame surrounding the door when the rotatable arcuate member is rotated to a locked position; and a stud extending from a side of the rotatable arcuate member, wherein the stud is configured to be received by the catch while the rotatable arcuate member is being rotated to the locked position such that the stud actuates the arm thereby moving the flag from a first position to a second position.
(5) In another embodiment, a flag mechanism for an aircraft door includes a first arm and a second arm each comprising a flag disposed on a first end and a catch disposed on a second end; a first cam operatively coupled to a first latching member and a second cam operatively coupled a second latching member, wherein the first and second latching members are each configured to be inserted into a frame surrounding the aircraft door when the first and second cams are rotated to a locked position, respectively; and a first stud extending from a side of the first cam and a second stud extending from a side of the second cam, wherein the first and second studs are each configured to be received by a respective catch of the first and second arms while the first and second cams are rotated to the locked position such that the first and second studs each actuate their respective arm thereby moving their respective flag from a first position to a second position.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES
(1) Embodiments of the invention are described in detail below with reference to the attached drawing figures, wherein:
(2) FIG. **1** illustrates a vehicle of some embodiments upon which a door may be included;
(3) FIG. **2**A illustrates an exterior view of a door in some embodiments;
(4) FIG. **2**B illustrates an interior view of the door of FIG. **2**A in some embodiments;
(5) FIG. **3** depicts some embodiments of a latching and locking system included in the door of FIG. **2**A;
(6) FIG. **4** illustrates some embodiments of a latch positioning mechanism of the door of FIG. **2**A in a pre-catch position;
(7) FIG. **5** depicts an interior view of the latch positioning mechanism of FIG. **4** in a released position, in some embodiments;
(8) FIG. **6** depicts a side perspective view of the latch positioning mechanism of FIG. **4** in a released position, in some embodiments;
(9) FIG. **7** illustrates a flag mechanism of some embodiments of the door of FIG. **2**A in an unlocked position;
(10) FIG. **8** illustrates the flag mechanism of FIG. **7** in some embodiments in a locked position;
(11) FIG. **9** depicts flags of the flag mechanism of FIG. **7**, in some embodiments; and
(12) FIG. **10** depicts a process-flow diagram illustrating a door latching and locking method using the system of FIGS. **2**A-**9**, in some embodiments.
(13) The drawing figures do not limit the invention to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the invention.
DETAILED DESCRIPTION
(14) The following detailed description references the accompanying drawings that illustrate specific embodiments in which the invention can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized, and changes can be made without departing from the scope of the invention. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the invention is defined only by the appended claims, along with the full scope of the equivalents to which such claims are entitled.
(15) In this description, references to “one embodiment,” “an embodiment,” or “embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment,” “an embodiment,” or “embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc. described in one embodiment may also be included in other embodiments, but is not necessarily included. Thus, the technology can include a variety of combinations and/or integrations of the embodiments described herein.
(16) Doors located on vehicles having high speeds and potentially having pressurized compartments, such as aircraft, require a latching and locking system that will both latch the door to maintain the door in a closed position and lock the latches to maintain the door in a locked position throughout the duration of the flight. These locking systems typically have latches that interface with the surrounding frame to hold the door in the closed position and locks that lock the latches to ensure they stay in the latched position. In some systems, this may necessitate a user to actuate each locking member individually. In yet other systems, a user may actuate one portion of the latching and locking system, such as a handle, which will cause actuation of the two or more

locking members at the same time. Regardless, safety protocols require the user confirm that the locking members are in in the locked position when the door is closed and locked by direct visual inspection. Typically, visual inspection is performed through windows located at the site of each locking member. Such placement of the window for visual inspection is inconvenient for some crew members, such as pilots, who may have aircraft components such as the seat, that block their view of the windows.
(17) Additionally, prior locking mechanisms may have a pre-catch mechanism that prevents the locking members from extending outwardly until the door is closed. However, these prior art pre-catch mechanisms typically have one or more biasing members (e.g., springs) that bias the locking members towards a pre-catch position (i.e., away from the airframe) even when the door is locked. Such a configuration is not ideal, as it is not desirable to have any mechanisms biasing the locking members away from a locked position when the door is to be locked.
(18) What is needed in the art is a door latching and locking mechanism having lock indication windows that are visible to crew members yet remote from the site of the locking mechanism. Additionally, the lock indication window should be placed where a crew member, such as a pilot or co-pilot, can easily view the lock indication window when seated in the cockpit. What is also needed in the art is a latch positioning mechanism that retains the locking members of the door in a pre-catch position (i.e., within the door) until the door is securely closed within the airframe, but does not bias the locking members towards the pre-catch position upon closing and locking of the door. Such a latch positioning mechanism would prevent the latching members from blocking the closing of the door while also not biasing the locking members towards the pre-catch position upon closing of the door.
(19) The flag mechanism utilizes bellcranks and flags to visually notify the crew member at the single lock indication window of the mechanical location of the locking members of the door. The latch positioning mechanism utilizes a series of a frame bumper, rods, bell cranks, and cams to keep the latching and locking mechanism in a pre-catch position while the door is open. Furthermore, the latch positioning mechanism is automatically released upon closing of the door. This release occurs without user input, and is initiated by contact between portions of the latch positioning mechanism and the frame. Releasing of the latch positioning mechanism prevents biasing of the locking members towards a pre-catch position when the door is closed and/or locked. Additionally, the latch positioning mechanism is automatically reengaged following unlocking of the locking mechanism and/or opening of the door, thereby placing the locking members back into a pre-catch position.
(20) FIG. **1** illustrates a vehicle **10** of some embodiments in which a door **100** may be located. The vehicle **10** may be a plane, helicopter, aircraft, boat, submarine, car, train, spacecraft, or any other vehicle that requires a latching and locking system within the door. While vehicle **10** may comprise any of the aforementioned options, vehicle **10** may be referred to as an aircraft for conciseness and clarity throughout the present disclosure. Vehicle **10** includes a frame **12** that surrounds the door **100**. In embodiments, the door **100** is rotatably coupled to frame **12** such that it pivots around an axis to open or close in comparison to the frame **12**. In embodiments, the frame **12** receives some or most of door **100** when in a closed position. As mentioned above, frame **12** may be configured to receive one or more locking members located in the door **100**, allowing the door **100** to be locked to the frame **12**. In some embodiments, frame **12** may be configured to receive two or more locking members. In yet other embodiments, frame **12** may be configured to receive two locking members.
(21) FIG. **2**A depicts an exterior view of door **100** while FIG. **2**B depicts an interior view of door **100** with some portions removed. As such, FIGS. **2**A and **2**B are best viewed together with the following description. In embodiments, door **100** is hingedly connected to frame **12** at a first side **102**. Further, a second side **104** of door **100** swings outwardly from frame **12** when door **100** is open. Further, second side **104** is received by, or within, frame **12** when door **100** is closed. While first side **102** is depicted as being connected at the fore side and second side **104** depicted as being

on the aft side, it is contemplated that first side **102** may be hingedly connected to any side of the vehicle **10**.
(22) As depicted in FIG. **2**B, door **100** includes a latching and locking system **200**. As will be described in greater detail below, latching and locking system **200** includes at least one latch. For example, in some embodiments, latching and locking system **200** includes a first latch **202** and a second latch **204**. As described above, frame **12** may be configured to receive first latch **202** and second latch **204** when door **100** is latched and/or locked into frame **12**. It is contemplated that frame **12** may have any variation of an indent, hole, lock, frame, etc. that is capable of receiving, or locking to, first latch **202** and second latch **204**. As will be discussed in greater detail, latching and locking system **200** may include an exterior handle **280** accessible on the exterior side of door **100** (e.g., see FIG. **2**A). Exterior handle **280** allows a user outside of aircraft **10** to unlock and open door **100**.
(23) FIG. **3** depicts latching and locking system **200**, in some embodiments, with certain components of door **100** removed for clarity. In embodiments, door **100** includes a latching and locking system **200**, a latch positioning mechanism **300** and a flag mechanism **400**. As will be discussed in greater detail below, latch positioning mechanism **300** is configured to maintain locking members **202**, **204** in a pre-catch position (i.e., within door **100**) when door **100** is open, thereby allowing door **100** to be pulled into frame **12** (i.e., closed). Flag mechanism **400** is configured to notify a user of latching and locking system **200** and door **100** of the mechanical location of first latch **202** and second latch **204**. While latch positioning mechanism **300** and flag mechanism **400** are described herein as separate systems within latching and locking system **200**, it is to be understood that latch positioning mechanism **300** and flag mechanism **400** are both operatively connected within latching and locking system **200** such that certain components are shared between latch positioning mechanism **300** and flag mechanism **400**.
(24) As illustrated in FIG. **3**, first latch **202** and second latch **204** are mechanically coupled within latching and locking system **200** via first latching rod **206** and second latching rod **208**, respectively. In some embodiments, first latch **202** is an extension of first latching rod **206** and second latch **204** is an extension of second latching rod **208**, the first and second locking members **202**, **204** extending into frame **12** when latching and locking system **200** is in a locked position. For example, actuation of first latching rod **206** will drive first latch **202** towards or away from frame **12**. Further, actuation of second latching rod **208** will drive second latch **204** towards or away from frame **12**. As will be discussed in greater detail below, first latching rod **206** is driven by rotation of a first cam **240** while second latching rod **208** is driven by rotation of a second cam **260**.
(25) First cam **240** and second cam **260** are for example rotatable arcuate members configure to be driven (e.g., rotated) by actuation of a handle **210**. For example, a portion of first cam **240** may be operatively connected to handle **210** via a first rod **214**. In another example, second cam **260** may be operatively connected to handle **210** via a second rod **218**, a rod bellcrank **220**, and a third rod **222**. In some embodiments, actuation of handle **210** may simultaneously actuate first rod **214** and second rod **218**, allowing for concurrent actuation of first cam **240** and second cam **260** via a single user interface. Concurrent operation of first latching rod **206** and second latching rod **208** allows for first latch **202** and second latch **204** to be actuated towards, or away, from frame **12** simultaneously.
(26) As will be described in greater detail below, rotation of first cam **240** and second cam **260** may also drive flag mechanism **400**. For example, rotation of first cam **240** may cause engagement of a catch **410***a* (see FIGS. **7** and **8**) with first cam **240**, thereby driving movement of a first flag **402**. In another example, rotation of second cam **260** may cause engagement of a catch **412***a* (see FIGS. **7** and **8**) with second cam **260**, thereby driving movement of a second flag **404**. As illustrated and described above, first cam **240** actuates first latch **202** and first flag **402**. Further, second cam **260** actuates second latch **204** and second flag **404**. Accordingly, the location/position of first flag **402** is indicative of the mechanical location of first latch **202** while the location/position of second flag

404 is indicative of the mechanical location of second latch **204**. As such, flag mechanism **400** allows a user to visually inspect and confirm the location/position of locking members **202**, **204**.
(27) Operative connection of the latch positioning mechanism **300** within the latching and locking system **200** is depicted in FIG. **3**, but will be described in greater detail below with reference to FIGS. **4-6**.
(28) FIG. **4** illustrates some embodiments of latch positioning mechanism **300** in the pre-catch position (e.g., when door **100** is open). FIGS. **5-6** illustrate some embodiments of latch positioning mechanism **300** in the released position (e.g., when door **100** is closed or received within frame **12**). Accordingly, FIGS. **4-6** are best viewed together for the following description.
(29) Latch positioning mechanism **300** includes a push rod **304** having a frame bumper **302** on an end (see FIG. **3**). In embodiments, the frame bumper **302** extends outwardly from door **100** such that when door **100** is shut within frame **12**, frame bumper **302** comes into contact with at least a portion of the frame **12**. For example, when door **100** is closed, frame bumper **302** may be displaced by contacting a portion of frame **12**. Accordingly, frame bumper **302** and push rod **304** will be passively displaced towards first side **102** of door **100**.
(30) As shown in FIG. **4**, in the pre-catch position, push rod **304** is mechanically coupled to a catch bellcrank **306**, which operatively engages a catch cam **308** at a first side **308***a*. As illustrated in FIG. **6**, first side **308***a* may be partially sloped towards catch bellcrank **306**. Such sloping of first side **308***a* aids in maintaining catch bellcrank **306** pressing against first side **308***a* in the pre-catch position. Catch cam **308** may be rotatable about pivot pin **211**. Further, in some embodiments, a portion of catch cam **308** may be attached to elastic member **310** (see FIGS. **5-6**). Elastic member **310** may bias first side **308***a* of catch cam **308** towards catch bellcrank **306**. Accordingly, when push rod **304** is actuated by contacting frame **12** at frame bumper **302**, catch bellcrank **306** is driven towards and down from first side **308***a*. Movement of catch bellcrank **306** towards and down from first side **308***a*, in combination with elastic member **310** driving first side **308***a* towards catch bellcrank **306**, causes rotation of catch cam **308** towards catch bellcrank **306** (e.g., in a clockwise direction as depicted in FIGS. **4-6**). In some embodiments, the shape of first side **308***a* is such that catch bellcrank **306** may rotate with little to no rotation of catch cam **308** as a roller **307** located on catch bellcrank **306** slides along first side **308***a*. In some embodiments, roller **307** may be configured to slidably contact first side **308***a* and second side **308***b* of catch cam **308** as latch positioning mechanism **300** transitions between the pre-catch position and the released position. Accordingly, as catch cam **308** is rotated about pivot pin **211**, roller **307** rides along a second side **308***b* of catch cam **308**. Further, in some embodiments catch bellcrank **306** includes a biasing member (e.g., a torsion spring) that biases roller **307** upwards (e.g., biases catch bellcrank **306** in a counter-clockwise direction as depicted in FIG. **4**).
(31) In embodiments, catch cam **308** is mechanically coupled to pivot pin **211**. Accordingly, rotation of catch cam **308** may additionally cause rotation of pivot pin **211**. Further, pivot pin **211** is mechanically coupled to handle **210** and seesaw **212**. As such, rotation of either of catch cam **308** or handle **210** may induce rotation of pivot pin **211** and subsequently seesaw **212**. As such, the latch positioning mechanism **300** prohibits rotation of handle **210** while in the pre-catch position as rotation of catch cam **308** is prevented via contact of catch bellcrank **306**. As illustrated between FIGS. **4** and **5**, rotation of seesaw **212** actuates first rod **214** and second rod **218**. For example, as illustrated, rotation of seesaw **212** in a clockwise direction around pivot pin **211** causes upward movement of first rod **214** and second rod **218**. As will be discussed in greater detail below, upward movement of both first rod **214** and second rod **218** cause rotation of first cam **240** and second cam **260**, respectively. As previously mentioned, rotation of first cam **240** and second cam **260** subsequently actuates first latch **202** and second latch **204** towards/into frame **12**. As catch cam **308** is mechanically coupled to pivot pin **211**, movement of locking members **202** and **204** is inhibited when catch cam **308** is prevented from rotating due to contact with catch bellcrank **306**. When catch bellcrank **306** and catch cam **308** are held in the pre-catch position, as shown in FIG. **4**,

locking members **202** and **204** are prevented from extending outwardly from door **100**. Because catch cam **308** is not released from the pre-catch position until frame bumper **302** contacts frame **12**, locking members **202** and **204** cannot extend outwardly from door **100** until catch cam **308** is released, as is shown in FIGS. **5** and **6**. Importantly, upon release of catch cam **308** and latch positioning mechanism **300**, there is no mechanical biasing of locking members **202**, **204** away from a locked position. It is also important to note that after the locking mechanism **200** is released from the pre-catch position by the latch positioning mechanism **300** and the elastic member **310** moves the locking mechanism **200** towards the latched and locked position, the door **100** is still not in a fully latched and locked configuration until further steps are taken by the user, such as actuating handle **210** which will be described in greater detail below.

(32) FIGS. **7-9** illustrate embodiments of the flag mechanism **400**. Specifically, FIG. **7** illustrates embodiments of flag mechanism **400** in an unlocked position. FIG. **8** illustrates embodiments of flag mechanism **400** in a locked position. FIG. **9** illustrates embodiments of the flags **402**, **404** of flag mechanism **400**. Accordingly, FIGS. **7-9** are best viewed together with the following description.

(33) As illustrated in FIG. **7**, when in the unlocked position, handle **210** has not been actuated from a first position (e.g., not horizontal) to a second position (e.g., substantially horizontal, see FIG. **8**). When handle **210** is in the first position, seesaw **212** is oriented such that first rod **214** and second rod **218** are substantially down, and first rod **214** has not yet driven rotation of first cam **240** around first cam pivot point **242**. Similarly, when handle **210** is in the first position, second rod **218** (via rod bellcrank **220** and third rod **222**) has not yet actuated rotation of second cam **260** around second cam pivot point **262**. As first cam **240** and second cam **260** are operatively and slidably coupled to first latch **202** and second latch **204** via first latching rod **206** and second latching rod **208**, respectively, when handle **210** is in the first position as shown in FIG. **7**, locking members **202**, **204** are not fully received by frame **12**.

(34) As described above, when the door **100** is actuated into or substantially in frame **12**, the latch positioning mechanism **300** will be actuated (i.e., released) due to contact of the frame bumper **302** with the surround frame **12**. In some embodiments, this will release catch cam **308** as discussed above. This allows a user to actuate components of the latching and locking system **200** and place the latching and locking system **200** in a latched and locked position, as will be discussed below.

(35) Following closing, door **100** is ready to be placed into a latched and locked position. Accordingly, a user within aircraft **10** may actuate handle **210**. As depicted in FIG. **8**, actuation of handle **210** in a downward direction to the second position may place the latching and locking system **200** in the locked position. As depicted, rotation of handle **210** around pivot pin **211** causes rotation of seesaw **212** in the same direction as handle **210**. Such rotation of seesaw **212** actuates both first rod **214** and second rod **218**. In some embodiments, both first rod **214** and second rod **218** are actuated in an upwards direction. As mentioned above, actuation of the first rod **214** and second rod **218** causes rotation of first cam **240** and second cam **260**. In some embodiments, biasing members located within latching and locking system **200** may maintain handle **210** in the second position. For example, elastic member **310** acting on catch cam **308** and subsequently pivot pin **211** biases handle **210** in the second position. Additionally, torsional biasing members may be disposed on rod bellcrank **220** and first cam **240**, which bias first rod **214** and second rod **218** in the upward, locked position. Accordingly, in some embodiments the force of one or more of the aforementioned biasing members must be overcome by a user of door **100** to actuate the handle **210** from the second position (i.e., latched and locked position) to the first position (i.e., unlatched and unlocked).

(36) Actuation of first rod **214** may cause rotation of first cam **240** around first cam pivot point **242**. As briefly mentioned above, first latching rod **206** is slidably and operatively coupled to first cam **240**. In embodiments, first latching rod **206** includes a pin **207** that is received within a first cam slot **244**. As depicted, first cam slot **244** is substantially rounded, having a first end **244***a* and a

second end **244***b*. In embodiments, the arc of first cam slot **244** is such that first end **244***a* is shorter distance from first cam pivot point **242** than second end **244***b* is from first cam pivot point **242**. Accordingly, as pin **207** slidably transitions within first cam slot **244** from first end **244***a* to second end **244***b*, pin **207** and first latching rod **206** are substantially actuated. In some embodiments, transition of pin **207** from first end **244***a* to second end **244***b* actuates first latching rod **206** horizontally, thereby causing first latch **202** to be received within surround frame **12**.

(37) Rotation of first cam **240** about first cam pivot point **242** further actuates first flag **402** of flag mechanism **400**. For example, in some embodiments, first cam **240** includes a stud **246**. Stud **246** extends transversely outward from first cam **240**. More specifically, stud **246** may extend from first cam **240** transversely in relation to the rotational plane of first cam **240**. In embodiments, stud **246** is configured to be “caught”, or received by catch **410***a* disposed on an end of a short arm **410**. Turning to FIG. **7**, stud **246** and catch **410***a* are depicted with a gap therebetween prior to rotation of first cam **240**. As illustrated in FIG. **8**, when first cam **240** is rotated (e.g., in a counterclockwise direction), stud **246** is received by catch **410***a* such that catch **410***a* and stud **246** are in direct contact with each other. The rotation of first cam **240** thereby causes rotation of short arm **410** and long arm **406** about a pivot point **405**. In embodiments, stud **246** is received within catch **410***a* prior to full rotation of first cam **240**. Accordingly, when stud **246** is received within catch **410***a*, first cam **240** continues to rotate, thereby actuating catch **410***a* and short arm **410**.

(38) In some embodiments, one or more of catch **410***a*, short arm **410**, and/or long arm **406** may be transiently maintained such that catch **410***a* is in the direct path of stud **246** as first cam **240** rotates. For example, catch **410***a*, short arm **410**, and/or long arm **406** may be held in place via a bracket, shelf, stopper, magnet, or any other means that may maintain the position of catch **410***a* while still allowing movement therefrom when stud **246** contacts catch **410***a*. As depicted in FIGS. **7** and **8** and described below, long arm **406** may be held in place via a stop bolt **407**.

(39) Actuation of short arm **410** further leads to actuation of long arm **406**. Such movement of long arm **406** actuates flag **402**. As depicted in FIGS. **7** and **7**, flag mechanism **400** includes a lock indication window **414**. In some embodiments, lock indication window **414** includes a first flag window **414***a* and a second flag window **414***b*. First flag window **414***a* may allow an onlooker from within aircraft **10** (e.g., a pilot in the cockpit) to view portions of first flag **402** while second flag window **414***b* may allow the same onlooker from within aircraft **10** to view portions of second flag **404**. Lock indication window **414**, therefore, is configured to allow for visual inspection of the mechanical state of components of latching and locking system **200** via the positioning of flags **402**, **404**. For example, as illustrated in FIG. **9**, first flag **402** comprises a first indicator **402***a* and a second indicator **402***b*. Similarly, second flag **404** includes a first indicator **404***a* and a second indicator **404***b*. First indicators **402***a*, **404***a* may be distinguished from second indicators **402***b*, **404***b* via any means commonly known to one skilled in the art. For example, first indicators **402***a*, **404***a* may be a different color, reflection, pattern, material, etc. to allow the onlooker to distinguish first indicators **402***a*, **404***a* from second indicators **402***b*, **404***b*. As can be seen in FIGS. **7-7**, first and second flag windows **414***a*, **414***b* may be configured such that an onlooker can only see either first indicator **402***a*, **404***a* or second indicator **402***b*, **404***b*. In the example depicted, first and second flag windows **414***a*, **414***b* are substantially circular in shape while first and second flags **402**, **404** are substantially elongated in shape. Further, in the example depicted, the long plane of the elongated shape is directed substantially along the same plane that flags **402**, **404** are adjusted along when actuated by first and second cams **240**, **260**. While circular and elongated shapes of flag windows **414***a*, **414***b* and flags **402**, **404**, respectively, are depicted, it is contemplated that windows **414***a*, **414***b* and flags **402**, **404** may consist of any shapes that allow for clear visual distinction between the position of first indicator **402***a*, **404***a* and second indicator **402***b*, **404***b* within windows **414***a*, **414***b*. As will be further discussed below, if first indicators **402***a*, **404***a* of flag mechanism **400** are viewed through flag windows **414***a*, **414***b*, this indicates to the onlooker that latching and locking system **200** is in an unlatched/unlocked position. Alternatively, if second

indicator **402***b*, **404***b* of flag mechanism **400** are viewed through flag windows **414***a*, **414***b*, this indicates to the onlooker that latching and locking system **200** is in a latched and locked position.

(40) Referring back to FIG. **8**, rotation of first cam **240** and stud **246** causes stud **246** to be received within and actuate catch **410***a*, which subsequently actuates short arm **410**, long arm **406**, and flag **402**. In some embodiments, such actuation of flag **402** changes the viewable portion of flag **402** through first flag window **414***a* from the first indicator **402***a* to the second indicator **402***b*. As mentioned above and is apparent from the preceding description, viewing of second indicator **402***b* through first flag window **414***a* indicates that first latching rod **206** and first latch **202** are actuated towards, and into surround frame **12**.

(41) Counter-rotation of first cam **240** about first cam pivot point **242** allows first flag **402** of flag mechanism **400** to be returned from the position shown in FIG. **8** back to the position shown in FIG. **7**. In embodiments, long arm **406** and short arm **410** are configured for gravity to pull long arm **406** downward to return flag **402** back to the FIG. **7** position. Long arm **406** is longer and heavier than short arm **410**; therefore, the long arm **406** is biased by gravity to rotate about pivot point **405** in a clockwise direction when catch **410***a* is free of stud **246**. A stop bolt **407** may be disposed beneath a portion of long arm **406** to support the long arm **406** when it is not engaged with first cam **240** cam via **246** stud. In some embodiments, a relatively weak magnet **417** is added to long arm **406** for holding long arm **406** against stop bolt **407** to prevent long arm **406** from bouncing up and down. The magnet **417** is weak enough to enable long arm **406** to rotate via stud **246** when first cam **240** is rotated. Alternatively, a weak spring may be disposed between stop bolt **407** and long arm **406** to transiently maintain long arm **406** downward against stop bolt **407** when not actuated via first cam **240**.

(42) Actuation of second latch **204** and second flag **404** may, in some embodiments, be similar to first latch **202** and first flag **402**, with slight variations. As mentioned above, actuation of handle **210** causes rotation of seesaw **212** around pivot pin **211**. Such movement of seesaw **212** causes second rod **218** to be displaced in an upwards direction. The opposing end of second rod **218** is operatively coupled to rod bellcrank **220** at a first end **220***a*. As illustrated in FIG. **8**, displacement of first end **220***a* via second rod **218** causes rod bellcrank **220** to rotate (e.g., in a counterclockwise direction). Rod bellcrank **220** is further operatively coupled to third rod **222** at a second end **220***b*. In embodiments, rotation of rod bellcrank **220** actuates third rod **222**.

(43) As mentioned above, third rod **222** is coupled to second cam **260**. As such, actuation of third rod **222** causes rotation of second cam **260** around second cam pivot point **262** (e.g., in a clockwise direction). Second cam **260** is slidably and operatively connected to second latching rod **208** (see FIGS. **3** and **7**). In some embodiments, second latching rod **208** includes pin **209** that is received by and slidably coupled to second cam slot **264**. Similar to first cam slot **244**, second cam slot **264** includes a first end **264***a* and a second end **264***b*. In embodiments, the arc of second cam slot **264** is such that second end **264***b* extends at a further distance from second cam pivot point **262** than does first end **264***a*. Accordingly, transition of pin **209** from first end **264***a* to second end **264***b* causes actuation of second latching rod **208**. Such actuation of second latching rod **208** via second cam **260** will subsequently drive second latch **204** into a portion of frame **12**, thereby placing it in a locked position, as shown in FIG. **8**.

(44) Similar to first cam **240**, second cam **260** includes a stud **266** extending outwardly therefrom. In embodiments, stud **266** extends in an outward direction from second cam **260** transversely in comparison to the rotational plane of second cam **260**. Stud **266** is configured to be “caught” or received within catch **412***a* located on an end of short arm **412**. Catch **412***a* may be placed such that stud **266** is received therein while second cam **260** has a further distance to rotate. In some embodiments, one or more of catch **412***a*, short arm **412**, and/or long arm **408** may be transiently maintained such that catch **412***a* is in the direct path of stud **266** as second cam **260** rotates (e.g., in the clockwise direction). For example, catch **412***a*, short arm **412**, and/or long arm **408** may be held in place via a bracket, shelf, stopper, magnet, or any other means that may maintain the position of

catch **412***a* while still allowing movement therefrom when stud **266** contacts catch **412***a*. Accordingly, further rotation of second cam **260** will actuate catch **412***a* and short arm **412**. Actuation of short arm **412** will further cause movement of long arm **408** and subsequently of second flag **404**.

(45) Movement of second flag **404** may subsequently alter the viewable portion of second flag **404** through second flag window **414***b*. For example, in an unlocked position (i.e., FIG. **7**) first indicator **404***a* may be viewable through second flag window **414***b*. However, when latching and locking system **200** transitions to the locked position (i.e., FIG. **8**), the position of second flag **404** may be adjusted such that second indicator **404***b* is viewable through second flag window **414***b*. Therefore, as mentioned above and is apparent from the preceding description, viewing of second indicator **404***b* through second flag window **414***b* indicates to the onlooker that second latch **204** has been mechanically adjusted into a locked position (e.g., driven into frame **12**).

(46) Similar to first cam **240**, counter-rotation of second cam **260** about second cam pivot point **262** allows second flag **404** to be returned from the position shown in FIG. **8** back to the position shown in FIG. **7**. In embodiments, long arm **408** and short arm **412** are configured for gravity to pull long arm **408** downward to return flag **404** back to the FIG. **7** position. Long arm **408** is longer and heavier than short arm **412**; therefore, the long arm **408** is biased by gravity to rotate about a pivot point **415** in a clockwise direction when catch **412***a* is free of stud **266**. A stop bolt **409** may be disposed beneath a portion of long arm **408** to support the long arm **408** when it is not engaged with second cam **260** via **266** stud. In some embodiments, a relatively weak magnet **419** is added to long arm **408** for holding long arm **408** against stop bolt **409** to prevent long arm **408** from bouncing up and down. The magnet **419** is weak enough to enable long arm **408** to rotate via stud **266** when second cam **260** is rotated. Alternatively, a weak spring may be disposed between stop bolt **409** and long arm **408** to transiently maintain long arm **408** downward against stop bolt **409** when not actuated via second cam **260**.

(47) Importantly, some or all of the steps described above may be reversed to move latching and locking system **200** and door **100** from the locked position (i.e., FIG. **8**) to the unlocked position (i.e., FIG. **7**). For example, handle **210** may be moved from the second position (e.g., FIG. **8**) to the first position (e.g., FIG. **7**). Such a movement causes seesaw **212** to rotate in a similar direction as handle **210**, thereby lowering both first rod **214** and second rod **218**. Lowering of first and second rods **214**, **218** causes rotation of first cam **240** and second cam **260** in a opposite direction than as previously described. This rotation would simultaneously release first latch **202** and second latch **204** while also releasing first flag **402** and second flag **404**. As such, an onlooker within aircraft **10** viewing first and second windows **414***a*, **414***b* would see first indicators **402***a*, **404***a*, thereby indicating that latching and locking system **200** has moved into an unlocked position.

(48) Also importantly, movement of handle **210** from the second position (e.g., FIG. **8**) to the first position (e.g., FIG. **7**) may move latch positioning mechanism **300** from the released position (e.g., FIGS. **5** and **6**) to the pre-catch position (e.g., FIG. **4**). For example, as discussed above, rotation of handle **210** from the second position to the first position will drive rotation of catch cam **308** in a counterclockwise direction via and around pivot pin **211**, as depicted. Accordingly, roller **307** will slide along second side **308***b* until second side **308***b* transitions into first side **308***a*. Once reached, catch bellcrank **306** will rotate (e.g., in a counterclockwise direction) such that roller **307** is substantially contacting first side **308***a* (e.g., FIG. **4**). As such, the latch positioning mechanism **300** is placed into the pre-catch position, thereby preventing movement of the first and second locking members **202**, **204** until the latch positioning mechanism **300** is released.

(49) As briefly mentioned above with reference to FIG. **2**A, latching and locking system **200** may further include an exterior handle **280** that allows an individual located outside of aircraft **10** to release latching and locking system **200** from the locked position. For example, exterior handle **280** may be operatively connected to rod **282** (see FIGS. **7-8**). As illustrated, rod **282** may be operatively connected to exterior handle **280** at a first end and connected to seesaw **212** at a second

end. Accordingly, a user may rotate, press, release, or otherwise actuate exterior handle **280** which in turn may drive movement of rod **282**. As illustrated, upward movement of rod **282** will actuate rotation of seesaw **212** away from the locked position (e.g., in the counterclockwise direction). Such rotation of seesaw **212** will subsequently drive first and second rods **214**, **218** in the downward direction, thereby driving locking members **202**, **204** out of frame **12**.

(50) It is noted that in the aforementioned descriptions of embodiments, directional descriptions such as clockwise, counterclockwise, horizontal, vertical, etc. are used merely for descriptive purposes only in relation to the illustrated figures. It is contemplated that other directions are used or understood to be used in certain circumstances. For example, clockwise and counterclockwise may be used interchangeably depending on which side of aircraft **10** door **100** is disposed or the arrangement of certain components within door **100**. Similarly, horizontal and vertical may be exchanged depending on the orientation of the components within door **100**. For example, it is contemplated that first latching rod **206** and first latch **202** may be arranged in a vertical direction rather than a horizontal direction. Furthermore, it is contemplated that second latching rod **208** and second latch **204** may be arranged in a horizontal direction rather than a vertical direction. Accordingly, any number of directions and descriptions of movement of components may be used without departing from the scope of the present disclosure.

(51) FIG. **10** depicts a process-flow diagram illustrating a door locking method **500** using the latching and locking system **200** of FIGS. **1-9**, in some embodiments. At step **502**, a door is actuated into, or substantially into, a surround frame. In an example of step **502**, a crew member, pilot, or user of airplane **10** actuates (e.g., pulls) door **100** into frame **12** (i.e., closes) such that the exterior of the door **100** is substantially flush with the exterior of frame **12**.

(52) At step **504**, a latch positioning mechanism is actuated and released. In an example of step **504**, latch positioning mechanism **300** is actuated when door **100** is received within frame **12**. For example, frame bumper **302** may contact an interior portion of surround frame **12**, thereby passively actuating components of latch positioning mechanism **300**. Upon contact of frame bumper **302**, push rod **304** may be displaced towards catch bellcrank **306**, thereby causing downward rotation of roller **307** located on catch bellcrank **306**. Such rotation of catch bellcrank **306** allows catch cam **308** to rotate towards catch bellcrank **306** (e.g., in a clockwise direction as depicted in FIG. **5**) due to elastic member **310** biasing catch bellcrank **306** in such a way. Rotation of catch bellcrank **306** repositions roller **307** from first side **308***a* to second side **308***b* of catch cam **308**, thereby moving the latch positioning mechanism **300** out of the pre-catch position and into the released position, thereby allowing latching and locking system **200** to be actuated in ways as previously described.

(53) In a step **506**, latching and locking members are actuated. In an example of step **506**, a user may actuate handle **210**, thereby driving a series of components of latching and locking system **200** to cause locking members **202**, **204** to be fully received within or coupled to frame **12**. For example, rotation of handle **210** into a second position causes seesaw **212** to rotate in the same direction (e.g., clockwise) around pivot pin **211**. Rotation of seesaw **212** drives first and second rods **214**, **218**. Such movement of first and second rods **214**, **218** causes rotation of first and second cams **240**, **260** around first cam pivot point **242** and second cam pivot point **262**, respectively. Once first and second cams **240**, **260** are fully rotated, pin **207** moves completely from first end **244***a* to second end **244***b* and pin **209** moves completely from first end **264***a* to second end **264***b*. Accordingly, first latching rod **206** and second latching rod **208** are fully actuated such that locking members **202**, **204** are fully received within surround frame **12**, thereby placing door **100** in a locked position.

(54) In a step **508**, a flag mechanism is actuated. In an example of step **508**, during rotation of first and second cams **240**, **260**, studs **246**, **266** are received within catches **410***a*, **412***a*, thereby actuating flags **402**, **404**. For example, as first cam **240** rotates about first cam pivot point **242**, stud **246** is received within catch **410***a*. This drives short arm **410** and long arm **406**, thereby displacing

first flag **402**. Displacement of first flag **402** may cause the viewable portion of first flag **402** through first flag window **414***a* to switch from first indicator **402***a* to second indicator **402***b*. In another example, as second cam **260** rotates about second cam pivot point **262**, stud **266** is received within catch **412***a*. This drives short arm **412** and long arm **408**, thereby displacing second flag **404**. Displacement of second flag **404** may cause the viewable portion of second flag **404** through second flag window **414***b* to switch from first indicator **404***a* to second indicator **404***b*.
(55) It is noted that while locking method **500** is depicted as sequential, two or more steps may take place simultaneously or mostly simultaneously. For example, step **502** may passively cause step **504** to occur at almost the same time by the frame bumper **302** coming into contact with frame **12**. Additionally, certain aspects of steps **506** and **508** may occur concurrently. For example, rotation of first and second cams **240**, **260** simultaneously drives first and second locking members **202**, **204** into frame **12** while also actuating components of flag mechanism **400**.
(56) As noted above, door **100** and latching and locking system **200** provide substantial benefits to users of an aircraft or other vehicle needing a safety lock mechanism within the doors. For example, latch positioning mechanism **300** provides a substantial benefit as it lacks any biasing members (e.g., springs) that bias any portion of latching and locking system **200** away from a locked position when door **100** is closed and/or locked. Instead, latch positioning mechanism **300** biases the locking members **202**, **204** towards a pre-catch position only while the door is open. Upon closure, the latch positioning mechanism **300** is automatically and passively released, thereby no longer holding latching and locking system **200** in the pre-catch position. Such a system is an improvement on prior art mechanisms, that continue to bias the locking members away from a locked position when the door is closed and/or locked. These prior art mechanisms may cause safety risks as such biasing away from the locked position is not preferable while the vehicle is operated. Furthermore, flag mechanism **400** provides substantial benefits to the user as they are capable of visually confirming the mechanical location of multiple locking members in one location (i.e., lock indication window **414**). For example, such capabilities may be especially advantageous for pilots who have a limited visual field of the inner doors when seated in the cockpit. Moreover, flag mechanism **400** is completely reliant upon mechanical systems to identify the position of the locking members. Therefore, flag mechanism **400** is not reliant upon electrical, wireless, or other non-mechanical systems which could, if broken, display a false-positive location of the locking members, thereby leading the crew member to believe the door is fully locked when it is in fact not locked.
(57) Although the invention has been described with reference to the embodiments illustrated in the attached drawing figures, it is noted that equivalents may be employed and substitutions made herein without departing from the scope of the invention as recited in the claims.
(58) Having thus described various embodiments of the invention, what is claimed as new and desired to be protected by Letters Patent includes the following:

Claims

1. A flag mechanism for a door of an aircraft, comprising: an arm comprising a flag disposed on a first end and a catch disposed on a second end; a rotatable arcuate member operatively coupled to a latching member, wherein the latching member is configured to be inserted into a frame surrounding the door when the rotatable arcuate member is rotated to a locked position; a stud extending from a side of the rotatable arcuate member, wherein the stud is configured to be received by the catch while the rotatable arcuate member is being rotated to the locked position such that the stud actuates the arm thereby moving the flag from a first position to a second position, wherein the stud is configured to be released by the catch, while the rotatable arcuate member is counter-rotated away from the locked position to an unlocked position, wherein the flag is configured to return to the first position upon the stud being released by the catch due to gravity,

and a magnet operatively coupled to the arm wherein the magnet is configured to provide a magnetic force that maintains the flag in the first position upon the stud being released by the catch.
2. The flag mechanism of claim 1, comprising a flag window, wherein the flag window has a viewable portion that is easily viewable to a crew member thereby enabling visual inspection of the flag by the crew member.
3. The flag mechanism of claim 2, wherein the flag is configured to provide an unlocked indicator that is viewable in the flag window to indicate that the door is unlocked when the flag is in the first position.
4. The flag mechanism of claim 2, wherein the flag is configured to provide a locked indicator that is viewable in the flag window to indicate that the door is locked when the flag is in the second position.
5. The flag mechanism of claim 2, wherein the flag has an elongated shape with an unlocked indicator on one end of the elongated shape and a locked indicator on an opposite end of the elongated shape.
6. The flag mechanism of claim 5, wherein the flag window has a round opening and a size of the round opening is such that only the unlocked indicator or the locked indicator is viewable through the flag window depending on the flag being in the first position or the second position, respectively.
7. A flag mechanism for an aircraft door, comprising: a first arm and a second arm each comprising a flag disposed on a first end and a catch disposed on a second end; a first cam operatively coupled to a first latching member and a second cam operatively coupled a second latching member, wherein the first and second latching members are each configured to be inserted into a frame surrounding the aircraft door when the first and second cams are rotated to a locked position, respectively; and a first stud extending from a side of the first cam and a second stud extending from a side of the second cam, wherein the first and second studs are each configured to be received by a respective catch of the first and second arms while the first and second cams are rotated to the locked position such that the first and second studs each actuate their respective arm thereby moving their respective flag from a first position to a second position.
8. The flag mechanism of claim 7, comprising a door handle operatively coupled to both the first and second cams such that rotation of the door handle between the locked and an unlocked position by a user rotates both the first and second cams thereby moving the first and second latching members between the latched and locked and unlatched and unlocked positions.
9. The flag mechanism of claim 8, wherein the first arm and the second arm are configured to both be actuated simultaneously by the first stud and the second stud as the door handle is moved from the unlocked position to the locked position.
10. The flag mechanism of claim 8, wherein the first stud and the second stud are both configured to be released by their respective catch while the first and second cams are counter-rotated away from the locked position to the unlocked position.
11. The flag mechanism of claim 8, wherein upon the first and second studs being released by their respective catch, each flag of the first and second arms is configured to return to the first position.
12. The flag mechanism of claim 11, wherein each of the first and second arms are configured to return to the first position via gravity.
13. The flag mechanism of claim 7, comprising a lock indication window, wherein the lock indication window comprises a first flag window co-located with a second flag window, wherein the first and second flag windows are both simultaneously viewable by a crew member seated inside an aircraft cockpit thereby enabling simultaneous visual inspection of both the first and second flag windows by the crew member for ensuring the first and second cams are in the locked position.
14. The flag mechanism of claim 13, wherein the first and second flags each comprise an elongated shape with an unlocked indicator on one end of the elongated shape and a locked indicator on an

opposite end of the elongated shape.

15. The flag mechanism of claim 14, wherein the first and second flag windows each has a round opening sized such that only the unlocked indicator or the locked indicator is viewable through a respective flag window depending on the first and second flags each being in the first position or the second position, respectively.

16. The flag mechanism of claim 15, wherein the lock indication window is configured for viewing both the first and second flags simultaneously to independently verify that both the first and second cams are in the locked position.